

Question

For a certain metal **X**, the oxide **A** of **X** and a common ternary acid mixed with its 1 : 1 ammonium salt are co-heated to 300°C . After washing with water and drying, compound **Y** is obtained, with a formula weight between 240 and 260, and contains no water of crystallization. If it is dissolved in dilute sulfuric acid, the solution color does not change significantly, but a precipitate is formed (reaction 1), while dissolving it in concentrated sulfuric acid yields a clear red solution. Heating compound **Y** to 340°C in an inert atmosphere results in a weight loss of 3.65%, yielding compound **B** (reaction 2); continuing to heat to 460°C results in another weight loss of 10.14%, yielding compound **C** (reaction 3); subsequently heating to 800°C does not result in significant weight loss, and compound **C** melts and completely transforms into compound **D**. The formula weight of **D** is 1.5 times that of **C**.

The chemical equation coefficients for all reactions are the simplest integers. Determine which of the following options is correct through reasoning.

- A. The +5 oxidation state is the stable oxidation state of metal **X**.
- B. **Y** has strong oxidizing properties
- C. The number of product species in reaction 1 is greater than the number of reactant species.
- D. The sum of stoichiometric coefficients of products in reaction 1 is greater than the sum of stoichiometric coefficients of reactants.
- E. The average relative molecular mass of the weight loss portion 3 is less than the average relative molecular mass of the weight loss portion 2.
- F. In all the compounds in this question, the metal elements have a total of 4 valence states.
- G. None of the above options is correct

Answer

Correct Answer: **C**

Detailed Explanation

Compound **Y** is soluble in concentrated sulfuric acid to obtain a clear red solution, possibly containing Mn compounds, ferricyanide, iodine, etc. Combined with other information in the question that there are no cyanide ions and iodine, it can be explained that the central metal is Mn; the composition of compound **Y** can also be analyzed. The common ternary acid is phosphoric acid, so **Y** may be ammonium phosphate or ammonium pyrophosphate. Combining the relative molecular mass range, ammonium pyrophosphate is a reasonable answer. At this time, the metal ion corresponds to the first transition period metal element. Combining the aforementioned chemical properties analysis, it can be determined that **X** is Mn. The +5 oxidation state is not a stable oxidation state of Mn, so option **A** is incorrect.

CHECKPOINT

1 PTS

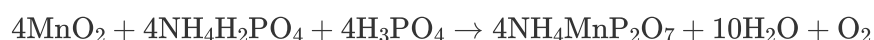
Metal **X** is Mn.

CHECKPOINT

0.5 PTS

+5 oxidation state is not a stable oxidation state of Mn, so option **A** is incorrect

The reaction equation for the formation of compound **Y**:

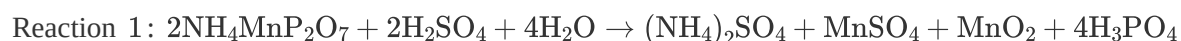


Mn in **Y** has a +3 oxidation state and does not have strong oxidizing properties, so option **B** is incorrect.

CHECKPOINT

1 PTS

Y is $\text{NH}_4\text{MnP}_2\text{O}_7$, in which Mn has a +3 oxidation state and does not have strong oxidizing properties, so option *B* is incorrect.



The number of species of products in reaction 1 is 4, the sum of the coefficients of products is 7, the number of species of reactants is 3, and the sum of the coefficients of reactants is 8, so option *C* is correct and option *D* is incorrect.

CHECKPOINT

1 PTS

The number of species of products in reaction 1 is 4, and the number of species of reactants is 3, the former is greater than the latter, so option *C* is correct.

CHECKPOINT

1 PTS

The sum of the coefficients of products in reaction 1 is 7, and the sum of the coefficients of reactants is 8, the former is less than the latter, so option *D* is incorrect.

Inert atmosphere heating weight loss process:

First step weight loss: $M_1 = w_1 \times H = 3.65 \times 10^{-2} \times 246.922 \text{ g} \cdot \text{mol}^{-1} = 9.01 \text{ g} \cdot \text{mol}^{-1}$, corresponding to 0.5 molecules of H_2O ;

CHECKPOINT

1 PTS

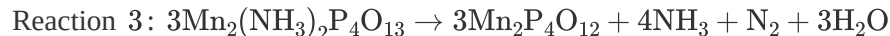
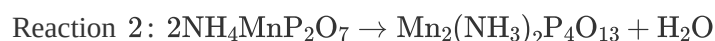
First step weight loss 9.01 g/mol , corresponding to $0.5 \text{ H}_2\text{O}$

Second step weight loss: $\text{H}_2 = w_2 \times \text{H} = 10.14 \times 10^{-2} \times 246.922 \text{ g} \cdot \text{mol}^{-1} = 25.04 \text{ g} \cdot \text{mol}^{-1}$, corresponding to 0.5 molecules of NH_3 and 0.5 molecules of NH_2OH (further decomposed into $\text{N}_2 \cdot \text{H}_2\text{O} \cdot \text{NH}_3$);

CHECKPOINT

1 PTS

Second step weight loss 25.04 g/mol



From the above, the average relative molecular mass of the weight loss portion of reaction 3:

$$m = (4 \times 17.034 + 28.02 + 3 \times 18.016)/8 = 18.78$$

The average relative molecular mass of the weight loss portion of reaction 3 is 18.78, and the average relative molecular mass of the weight loss portion of reaction 2 is 18.02, so option *E* is incorrect.

In the compounds involved in the entire question, there are three oxidation states of Mn: +2 oxidation state ($\text{Mn}_2\text{P}_4\text{O}_{12}$), +3 oxidation state ($\text{NH}_4\text{MnP}_2\text{O}_7$), and +4 oxidation state (MnO_2), so option *F* is incorrect.

CHECKPOINT

0.5 PTS

The average relative molecular mass of the weight loss portion of reaction 3 is 18.78, and the average relative molecular mass of the weight loss portion of reaction 2 is 18.02, so option *E* is incorrect.

CHECKPOINT

1 PTS

In the compounds involved in the entire question, there are three oxidation states of Mn: +2 oxidation state ($\text{Mn}_2\text{P}_4\text{O}_{12}$), +3 oxidation state ($\text{NH}_4\text{MnP}_2\text{O}_7$), and +4 oxidation state (MnO_2), so option *F* is incorrect.